Continuum Mechanics
class # 18
10/20/2026

Entropy Production

We have already proven the Clausius – Duhem inequality in Eulerian form

$$\rho\dot{\eta} = \frac{\rho r}{\theta} - \nabla^x \cdot \left(\frac{\vec{q}}{\theta}\right) + \sigma, \tag{5.22}$$

where σ is the entropy production term:

$$\sigma(\vec{x},t) \geq 0, \quad \forall\, \vec{x} \in B_t, \quad \forall\, t \geq 0.$$

The book uses a slightly different form. To get this form, multiply (5.22) by θ:

$$\begin{aligned} \theta\rho\dot{\eta} &= \rho r - \theta\left\{\frac{1}{\theta}\nabla^x\cdot\vec{q} + \vec{q}\cdot\nabla^x\left(\frac{1}{\theta}\right)\right\} + \theta\sigma \\ &= \rho r - \nabla^x\cdot\vec{q} + \theta\vec{q}\cdot\frac{1}{\theta^2}\nabla^x\theta + \theta\sigma \qquad (5.23) \\ &= \rho r - \nabla^x\cdot\vec{q} + \left(\frac{\vec{q}}{\theta}\right)\cdot\nabla^x\theta + \theta\sigma. \end{aligned}$$

Define

$$\delta := \theta\rho\dot{\eta} - \rho r + \nabla^x\cdot\vec{q}.$$

This quantity, δ, is called the <u>internal dissipation density</u>. Using this new quantity

$$\delta - \left(\frac{\vec{q}}{\theta}\right)\cdot\nabla^x\theta = \theta\sigma \geq 0. \tag{5.24}$$

Definition: Suppose that ϕ is the internal energy density field per unit mass, η is the entropy density field per unit mass, and θ is the temperature field. The quantity

(5.25) $$\psi(\vec{x},t) = \phi(\vec{x},t) - \theta(\vec{x},t)\,\eta(\vec{x},t)$$

is called the **free energy density field** per unit mass.

Remark: In the classical theory of homogeneous systems undergoing reversible processes, free energy is that portion of the internal energy available for performing work at constant temperature.

Result (5.12): Let $\vec{\varphi}: B \times [0,\infty) \to \mathbb{E}^3$ be a motion of a continuum body with spatial free energy density $\psi(\vec{x},t)$. Then

$$\rho\dot{\psi} = S : L - \rho\eta\dot{\theta} - \left(\frac{\vec{q}}{\theta}\right)\cdot\nabla^x\theta - \theta\sigma$$

Proof: From (5.25)

$$\dot{\psi} = \dot{\phi} - (\dot{\theta\eta})$$

$$(\dot{\theta\eta}) = \frac{\partial}{\partial t}(\theta\eta) + \nabla^x(\theta\eta)\cdot\vec{v}$$

$$= \frac{\partial\theta}{\partial t}\eta + \theta\frac{\partial\eta}{\partial t} + \eta\nabla^x\theta\cdot\vec{v} + \theta\nabla^x\eta\cdot\vec{v}$$

$$= \dot{\theta}\eta + \theta\dot{\eta}$$

So

$$\theta\dot{\eta} = \dot{\phi} - \dot{\theta}\eta - \dot{\psi}$$

Using (5.23) and Result (5.10)

$$\theta\rho\dot{\eta} = \rho r - \nabla^x \cdot \vec{q} + \left(\frac{\vec{q}}{\theta}\right) \cdot \nabla^x \theta + \theta\sigma.$$

$$= \rho\dot{\phi} - \rho\dot{\theta}\eta - \rho\dot{\psi}$$

$$= S:L - \nabla^x \cdot \vec{q} + \rho r - \rho\dot{\theta}\eta - \rho\dot{\psi}$$

Thus,

$$\left(\frac{\vec{q}}{\theta}\right) \cdot \nabla^x \theta + \theta\sigma = S:L - \rho\dot{\theta}\eta - \rho\dot{\psi}$$

Rearranging terms

$$\rho\dot{\psi} = S:L - \left(\frac{\vec{q}}{\theta}\right) \cdot \nabla^x \theta - \rho\dot{\theta}\eta - \theta\sigma \quad \text{///}$$

Corollary: Since $\sigma \geq 0$ in B_t for all $t \geq 0$,

$$\rho\dot{\psi} \leq S:L - \left(\frac{\vec{q}}{\theta}\right) \cdot \nabla^x \theta - \rho\eta\dot{\theta}.$$

Summary of Eulerian Variables

In the Eulerian description there are 22 unknowns to be determined

$[\vec{\phi}(\vec{x},t)]_i$	3	components of motion
$[\vec{v}(\vec{x},t)]_i$	3	components of velocity
$\rho(\vec{x},t)$	1	mass density
$[S(\vec{x},t)]_{ij}$	9	Components of stress
$\theta(\vec{x},t)$	1	temperature
$[\vec{q}(\vec{x},t)]_i$	3	components of heat flux
$\phi(\vec{x},t)$	1	internal energy density
$\eta(\vec{x},t)$	1	entropy density

To determine the unknowns we have, so far, 11 equations

$[v_i]_m = \frac{\partial}{\partial t}\phi_i$	3	Kinematical
$\frac{\partial \rho}{\partial t} + \nabla^x \cdot (\rho \vec{v}) = 0$	1	mass
$[\rho \dot{\vec{v}}]_i = [\nabla^x \cdot S + \rho \vec{b}]_i$	3	linear momentum
$[S]_{ij} = [S]_{ji}$	3	angular momentum
$\rho \dot{\phi} = S : L - \nabla^x \cdot \vec{q} + \rho r$	1	energy

We need constitutive laws to close the system so that it can be solved.

Lagrangian Local Conservation of Mass

Result (5.13): Let $\vec{\phi} : B \times [0,\infty) \to \mathbb{E}^3$ be a motion of a continuum body with associated deformation gradient field $F(\vec{X},t)$ and spatial mass density field $\rho(\vec{x},t)$. Let $\rho_0(\vec{X})$ be the mass density field in the reference configuration B. Then Axiom of Mass conservation, ie,

$$\text{mass}[\mathcal{Q}_t] = \text{mass}[\mathcal{Q}_0],$$

implies that

(5.18)
$$\rho_m(\vec{X},t) \det F(\vec{X},t) = \rho_0(X), \qquad \forall\, X \in B,\ t \geq 0,$$

where $\rho_m(\vec{X},t)$ is the material description of the spatial density $\rho(\vec{x},t)$.

Proof: Equation (5.18) was proven earlier. ///

Lagrangian Local Conservation of Momentum

Definition (5.14): Let $S(\vec{x},t)$ be the Cauchy stress field in B_t associated with a motion $\vec{\varphi} : B \times [0,\infty) \to \mathbb{E}^3$. The first Piola-Kirchoff stress tensor associated with $\vec{\varphi}$ is defined as

$$P(\vec{X},t) := \det(F(\vec{X},t))\, S_m(\vec{X},t)\, F(\vec{X},t)^{-T}$$

for all $X \in B$, for all $t \geq 0$.

The <u>second Piola-Kirchoff stress tensor</u> associated with $\vec{\varphi}$ is

$$\Sigma(\vec{X},t) := F(X,t)^{-1} P(\vec{X},t)$$

for all $\vec{X} \in B$, for all $t \geq 0$.

<u>Result</u> (5.15): Let $\vec{\varphi} : B \times [0,\infty) \to \mathbb{E}^3$ be a motion of a continuum body and let $\rho_0(\vec{X})$ denote the mass density field in the reference configuration B. Then, the balance of linear momentum implies

$$\rho_0 \ddot{\vec{\varphi}} = \nabla^X \cdot P + \rho_0 \vec{b}_m, \quad \forall \vec{X} \in B, \ t \geq 0,$$

where $P(\vec{X},t)$ is the first Piola-Kirchoff stress tensor and $\vec{b}_m(\vec{X},t)$ is the material description of the spatial body force field $\vec{b}(\vec{x},t)$.

<u>Proof</u>: As stated in Axiom (5.2), the balance law for linear angular momentum for an arbitrary open set $\Omega_t \subseteq B_t$ is

$$\frac{d}{dt} \vec{l}[\Omega_t] = \vec{r}[\Omega_t]$$

where $\vec{l}[\Omega_t]$ is the linear momentum of Ω_t defined by

$$\vec{l}[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t) \vec{v}(\vec{x},t) \, dV_x$$

and the resultant external force on Ω_t is

$$\vec{r}[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t) b(\vec{x},t) \, dV_x + \int_{\partial\Omega_t} S(\vec{x},t) \vec{n}(\vec{x}) \, dA_x .$$

Performing a change of variables in $\vec{l}[\Omega_t]$, we have

$$\vec{l}[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t)\vec{v}(\vec{x},t)\,dV_x$$

$$= \int_{\Omega_0} \rho(\vec{\varphi}(\vec{X},t),t)\,\vec{v}(\vec{\varphi}(\vec{X},t),t)\,J(\vec{X},t)\,dV_X$$

$$= \int_{\Omega_0} \rho_0(\vec{X})\,\dot{\vec{\varphi}}(\vec{X},t)\,dV_X,$$

using Result (5.13), (Equation (5.18)). Performing a change of variables in $\vec{r}[\Omega_t]$, we have

$$\vec{r}[\Omega_t] = \int_{\partial\Omega_t} S(\vec{x},t)\vec{w}(\vec{x})\,dA_x + \int_{\Omega_t} \rho(\vec{x},t)\vec{b}(\vec{x},t)\,dV_x$$

$$= \int_{\partial\Omega_0} \det(F(\vec{X},t))\,S(\vec{\varphi}(\vec{X},t),t)\,F(\vec{X},t)^{-T}N(\vec{X})\,dA_X$$

$$+ \int_{\Omega_0} \det(F(\vec{X},t))\,\rho(\vec{\varphi}(\vec{X},t),t)\,\vec{b}(\vec{\varphi}(\vec{X},t),t)\,dV_X$$

$$= \int_{\partial\Omega_0} P(\vec{X},t)\,N(X)\,dA_X + \int_{\Omega_0} \rho_0(\vec{X})\,\vec{b}_m(\vec{X},t)\,dV_X$$

Now,

$$\frac{d}{dt}\vec{l}[\Omega_t] = \int_{\Omega_0} \rho_0(\vec{X})\,\ddot{\vec{\varphi}}(\vec{X},t)\,dt.$$

Using the Divergence Theorem

$$\vec{r}[\Omega_t] = \int_{\Omega_0} \left(\nabla^X \cdot P + \rho_0(\vec{X}) \vec{b}_m(\vec{X}, t) \right) dV_X .$$

Using the localization Theorem, we have,

$$\rho_0(\vec{X}) \ddot{\vec{\phi}}(\vec{X}, t) = \nabla^X \cdot P + \rho_0(\vec{X}) \vec{b}_m(\vec{X}, t),$$

for all $X \in B$, for all $t \geq 0$. ///

Result (5.16): Let $\vec{\varphi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of continuum body associated with the deformation gradient $F(\vec{X}, t)$. Then, the balance of angular momentum requires

$$PF^T = FP^T, \quad \forall \vec{X} \in B, \ \forall t \geq 0,$$

or, equivalently,

$$\Sigma^T = \Sigma, \quad \forall \vec{X} \in B, \ \forall t \geq 0,$$

where P is the first Piola Kirchoff stress tensor and Σ is the second.

Proof: The result follows by applying Result (5.8) and Definition (5.14). ///

The First Law of Thermodynamics

Result (5.17): Let $\vec{\varphi} : B \times [0, \infty) \to \mathbb{E}^3$ be a motion of a continuum body with deformation gradient $F(\vec{X}, t)$ and first Piola-Kirchoff stress tensor $P(\vec{X}, t)$.

Let $\Omega_t \subseteq B_t$ be arbitrary. Then

$$\frac{d}{dt} K[\Omega_t] + \int_{\Omega} P : \dot{F} \, dV_x = \mathcal{P}[\Omega_t]$$

for all $t \geq 0$, where $K[\Omega_t]$ is the kinetic energy and $\mathcal{P}[\Omega_t]$ is the power of external forces on Ω_t. In view of the definition

$$\mathcal{W}[\Omega_t] = \mathcal{P}[\Omega_t] - \frac{d}{dt} K[\Omega_t], \tag{5.14}$$

it follows that

$$\mathcal{W}[\Omega_t] = \int_{\Omega_0} P : \dot{F} \, dV_x.$$

Proof: By definition

$$K[\Omega_t] = \int_{\Omega_t} \frac{1}{2} \rho(\vec{x},t) |\vec{v}(\vec{x},t)|^2 \, dV_x$$

and

$$\mathcal{P}[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t) \vec{b}(\vec{x},t) \cdot \vec{v}(\vec{x},t) \, dV_x + \int_{\partial\Omega_t} S(\vec{x},t)^T \vec{v}(\vec{x},t) \cdot \vec{n}(\vec{x}) \, dA_x,$$

where we have used

$$S^T \vec{v} \cdot \vec{n} = \vec{v} \cdot S\vec{n}.$$

Performing a change of variables on $K[\Omega_t]$ and using Result (5.13), we have

$$K[\Omega_t] = \int_{\Omega_0} \frac{1}{2} \rho_0(\vec{x}) |\dot{\vec{\Phi}}(\vec{x},t)|^2 \, dV_x \tag{5.30}$$

Using a change of variables for $P[\Omega_t]$ we have

$$P[\Omega_t] = \int_{\Omega_0} \rho_0 \vec{b}_m \cdot \vec{v}_m \, dV_X + \int_{\partial\Omega_0} \det(F) \vec{v}_m \cdot S_m F^{-T} N \, dA_X$$

$$= \int_{\Omega_0} \rho_0 \vec{b}_m \cdot \dot{\vec{\varphi}} \, dV_X + \int_{\partial\Omega_0} (P^T \dot{\vec{\varphi}}) \cdot \vec{N} \, dA_X$$

By the Divergence Theorem and Result (2.11),

$$P[\Omega_t] = \int_{\Omega_0} \{ \rho_0 \vec{b}_m \cdot \dot{\vec{\varphi}} + \nabla^X \cdot (P^T \dot{\vec{\varphi}}) \} \, dV_X$$

(5.31)

$$= \int_{\Omega_0} \{ (\nabla^X \cdot P + \rho_0 \vec{b}_m) \cdot \dot{\vec{\varphi}} + P : \nabla^X \dot{\vec{\varphi}} \} \, dV_X$$

Taking the time derivative of (5.30) and using Result (5.15) we have

$$\frac{d}{dt} K[\Omega_t] = \int_{\Omega} \rho_0 \dot{\vec{\varphi}} \ddot{\vec{\varphi}} \, dV_X$$

$$= \int_{\Omega} \dot{\vec{\varphi}} \cdot (\nabla^X \cdot P + \rho_0 \vec{b}_m) \, dV_X .$$

Now, observe that

$$\nabla^X \dot{\vec{\varphi}}(\vec{X}, t) = \dot{F}(\vec{X}, t),$$

So that the result follows. ///

Result (5.18): Let $\vec{\varphi} : \mathcal{B} \times [0,\infty) \to \mathbb{E}^3$ be a motion of a continuous body with associated material internal energy field $\mathfrak{E}(\vec{X},t)$,

$$\mathfrak{E}(\vec{X},t) = \phi_m(\vec{X},t) = \phi(\vec{\varphi}(\vec{X},t),t),$$

material heat flux $\vec{Q}$,

$$\vec{Q}(\vec{X},t) := \det(F(\vec{X},t))\, F(\vec{X},t)^{-1} \vec{q}_m(\vec{X},t),$$

and material heat supply R,

$$R(\vec{X},t) = r_m(\vec{X},t) = r(\vec{\varphi}(\vec{X},t),t).$$

Then, the First law of Thermodynamics implies

$$\rho_0 \dot{\mathfrak{E}} = P : \dot{F} - \nabla^X \cdot \vec{Q} + \rho_0 R,$$

for all $\vec{X} \in \mathcal{B}$ and all $t \geq 0$.

Proof: The First law states

$$\frac{d}{dt} U[\Omega_t] = Q[\Omega_t] + W[\Omega_t], \tag{5.32}$$

where $U[\Omega_t]$ is the internal energy of Ω_t, defined as

$$U[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t)\, \phi(\vec{x},t)\, dV_x,$$

$Q[\Omega_t]$ is the net heating of Ω_t, defined as

$$Q[\Omega_t] = \int_{\Omega_t} \rho(\vec{x},t)\, r(\vec{x},t)\, dV_x - \int_{\partial\Omega_t} \vec{q}(\vec{x},t) \cdot \vec{n}(\vec{x})\, dA_x$$

and $\mathcal{W}[\Omega_t]$ is the net working of external forces on Ω_t.

By a change of variables

(5.33) $$U[\Omega_t] = \int_{\Omega_0} \rho_0(\vec{X})\, \Phi(\vec{X}, t)\, dV_X$$

Similarly,

$$Q[\Omega_t] = \int_{\Omega_0} \rho_0 r_m\, dV_X - \int_{\partial\Omega_0} \det(F)\, \vec{q}_m \cdot F^{-1} N\, dA_X$$

(5.35)

$$= \int_{\Omega_0} \rho_0 R\, dV_X - \int_{\partial\Omega_0} Q \cdot \vec{N}\, dA_X .$$

Taking the time derivative of (5.33) and using (5.32),

$$\frac{d}{dt} U[\Omega_t] = \int_{\Omega_0} \rho_0(X)\, \dot{\Phi}(\vec{X}, t)\, dV_X$$

$$= \int_{\Omega_0} \rho_0 R\, dV_X - \int_{\partial\Omega_0} \vec{Q} \cdot N\, dA_X$$

$$+ \int_{\Omega_0} P : \dot{F}\, dV_X .$$

Using the Divergence Theorem and Localization, the Result follows. ///